Chapter 29

Campbell Biology, AP Edition, 12e

Lush landscape filled with ferns, moss, and trees.

Chapter 29

Plant Diversity I: How Plants Colonized Land

Figure 29.1

For much of Earth's history, the land surface was largely lifeless. Some prokaryotes lived on land 3.2 billion years ago, but it was only within the last 500 million years that small plants, fungi, and animals joined them ashore. Finally, by 385 million years ago, the first forests appeared (but with different species than those of today).

What were the major developments in the evolution of plants?

Plants originated from green algae about 470 million years ago. By 425 million years ago, some early plants had traits facilitating life on land. These traits include reproductive structures, photosynthetic branches, and structures that anchor the plant to the soil. Over time, early plants gave rise to a rich diversity of plants, including the following: nonvascular plants (such as mosses), seedless vascular plants (such as ferns), and seed plants (such as cone-bearing plants and flowering plants).

Key Concepts

- 29.1**Plants evolved from green algae******
- 29.2**Mosses and other nonvascular plants have life cycles dominated by gametophytes******
- 29.3**Ferns and other seedless vascular plants were the first plants to grow tall******

Advanced placement edition

As organisms colonized land, the changed environment acted as a selective mechanism for particular traits that ensured reproductive success **(BIG IDEA 1)** and led to differentiation in the use of energy as organisms exchanged matter with the environment **(BIG IDEA 2)** and interacted with each other, recycling matter **(BIG IDEA 4)**.

Study Tip

Make a table: As you read the chapter, make a table of adaptations for life on land that evolved in plants after they diverged from their close aquatic relatives, a group of charophyte algae.

A partially filled table lists adaptation, description, and how it facilitates life on land.

The table is as follows:

Adaptation	Description	How it facilitates life on land
Cuticle	Waxy covering on outer body surface	Reduces desiccation
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Get Ready for This Chapter

In this chapter, you will learn about the colonization of land by plants, focusing on the first 100 million years of plant evolution. Before beginning this chapter, you should be able to do the following:

- Explain the process of photosynthesis (see Concept 10.2, Figure 10.5, and Figure 10.21).
- Define and apply the terms *haploid* and *diploid* (see Concept 13.2).
- Identify the algal group from which plants arose (see Figure 28.5).

The following questions provide a quick check of your basic knowledge in these areas.

1. Which of the following chemical equations summarizes the reactions in photosynthesis?

```
    Light energy converts carbon dioxide and water to
sugars and oxygen.
```

That's correct.

Correct! If you had trouble with this question, review the following material: Photosynthesis begins when light energy is used to split molecules of water. The process releases oxygen to the atmosphere and is used to convert light energy to the chemical energy of ATP and NADPH. Those molecules are then used (in the Calvin cycle) to convert carbon dioxide to sugars, such as glucose.

```
        Light energy converts oxygen and sugars to water and
carbon dioxide.
```

That's incorrect. Try again.

Incorrect. Review the following material: Photosynthesis begins when light energy is used to split molecules of water. The process releases oxygen to the atmosphere and is used to convert light energy to the chemical energy of ATP and NADPH. Those molecules are then used (in the Calvin cycle) to convert carbon dioxide to sugars, such as glucose.

```
        Sugars and oxygen convert to light energy, carbon
dioxide, and water.
```

That's incorrect. Try again.

Incorrect. Review the following material: Photosynthesis begins when light energy is used to split molecules of water. The process releases oxygen to the atmosphere and is used to convert light energy to the chemical energy of ATP and NADPH. Those molecules are then used (in the Calvin cycle) to convert carbon dioxide to sugars, such as glucose.

```
        Carbon dioxide and oxygen convert to light energy,
sugars, and water.
```

That's incorrect. Try again.

Incorrect. Review the following material: Photosynthesis begins when light energy is used to split molecules of water. The process releases oxygen to the atmosphere and is used to convert light energy to the chemical energy of ATP and NADPH. Those molecules are then used (in the Calvin cycle) to convert carbon dioxide to sugars, such as glucose.

2. If the diploid chromosome number of a sexually-reproducing species is 58, which of the following statements is true?

The haploid chromosome number is 116.

That's incorrect. Try again.

Incorrect. Review the following material: Each somatic cell of a sexually-reproducing species contains pairs of homologous chromosomes. One chromosome of each pair is inherited from each parent. For example, in humans, the 46 chromosomes in the somatic cells are actually two sets of 23 chromosomes—a maternal set (from the mother) and a paternal set (from the father). The number of chromosomes in a single set is represented by n. Any cell with two chromosome sets has a diploid number of chromosomes, abbreviated 2n. For humans, the diploid number is 46 (2n=46). Unlike somatic cells, gametes contain a single set of chromosomes. Such cells have a haploid number of chromosomes (n). For humans, the haploid number is 23 (n=23).

Each gamete contains 58 chromosomes.

That's incorrect. Try again.

Incorrect. Review the following material: Each somatic cell of a sexually-reproducing species contains pairs of homologous chromosomes. One chromosome of each pair is inherited from each parent. For example, in humans, the 46 chromosomes in the somatic cells are actually two sets of 23 chromosomes—a maternal set (from the mother) and a paternal set (from the father). The number of chromosomes in a single set is represented by n. Any cell with two chromosome sets has a diploid number of chromosomes, abbreviated 2n. For humans, the diploid number is 46 (2n=46). Unlike somatic cells, gametes contain a single set of chromosomes. Such cells have a haploid number of chromosomes (n). For humans, the haploid number is 23(n=23).

There are 58 pairs of homologous chromosomes in each cell.

That's incorrect. Try again.

Incorrect. Review the following material: Each somatic cell of a sexually-reproducing species contains pairs of homologous chromosomes. One chromosome of each pair is inherited from each parent. For example, in humans, the 46 chromosomes in the somatic cells are actually two sets of 23 chromosomes—a maternal set (from the mother) and a paternal set (from the father). The number of chromosomes in a single set is represented by n. Any cell with two chromosome sets has a diploid number of chromosomes, abbreviated 2n. For humans, the diploid number is 46(2n=46). Unlike somatic cells, gametes contain a single set of chromosomes. Such cells have a haploid number of chromosomes (n). For humans, the haploid number is 23(n=23).

The haploid chromosome number is 29.

That's correct.

Correct! If you had trouble with this question, review the following material: Each somatic cell of a sexually-reproducing species contains pairs of homologous chromosomes. One chromosome of each pair is inherited from each parent. For example, in humans, the 46 chromosomes in the somatic cells are actually two sets of 23 chromosomes—a maternal set (from the mother) and a paternal set (from the father). The number of chromosomes in a single set is represented by n. Any cell with two chromosome sets has a diploid number of chromosomes, abbreviated 2n. For humans, the diploid number is 46(2n=46). Unlike somatic cells, gametes contain a single set of chromosomes. Such cells have a haploid number of chromosomes (n). For humans, the haploid number is 23(n=23).

3. Plants are members of the eukaryotic supergroup _____, and their closest relatives are _____.

```
Unikonta; charophyte green algae.
```

That's incorrect. Try again.

Incorrect. Review the following material: There are four eukaryotic supergroups: Excavata, SAR, Archaeplastida, and Unikonta. Plants are members of Archaeplastida. Charophyte green algae are the sister group to plants; thus, the closest relatives of plants are charophyte green algae.

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Archaeplastida; charophyte green algae.
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That's correct.

Correct! If you had trouble with this question, review the following material: There are four eukaryotic supergroups: Excavata, SAR, Archaeplastida, and Unikonta. Plants are members of Archaeplastida. Charophyte green algae are the sister group to plants; thus, the closest relatives of plants are charophyte green algae.

```
Archaeplastida; chlorophyte green algae.
```

That's incorrect. Try again.

Incorrect. Review the following material: There are four eukaryotic supergroups: Excavata, SAR, Archaeplastida, and Unikonta. Plants are members of Archaeplastida. Charophyte green algae are the sister group to plants; thus, the closest relatives of plants are charophyte green algae.

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Unikonta; chlorophyte green algae.
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That's incorrect. Try again.

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Learning Objectives

After studying this chapter, you should be able to do the following:

- 29.1Identify key derived characters of plants.
- 29.2Characterize the life cycles of nonvascular plants.
- 29.3Describe characteristics and reproductive processes of seedless vascular plants.

Concept 29.1: Plants evolved from green algae

Today, there are more than 325,000 known plant species. Although a few plant species returned to aquatic habitats during their evolution, most present-day plants live on land. In this text, we distinguish plants from algae, which are photosynthetic protists.

Plants enabled other life-forms to survive on land. For example, plants supply oxygen and are a key source of food for terrestrial animals. Also, by their very presence, plants such as the trees of a forest physically create the habitats required by animals and many other organisms.

As you read in Concept 28.4, green algae called charophytes are the closest relatives of plants. We'll begin with a closer look at the evidence for this relationship.

Evidence of Algal Ancestry

Many key traits of plants are found in some algae. For example, plants are multicellular, eukaryotic, photosynthetic autotrophs, as are brown, red, and certain green algae. Plants have cell walls made of cellulose, and so do green algae, dinoflagellates, and brown algae. And chloroplasts with chlorophylls a and b are present in green algae, euglenids, and a few dinoflagellates, as well as in plants.

However, the charophytes are the only present-day algae that share certain distinctive traits with plants, suggesting that they are the closest living relatives of plants. For example, the cells of both plants and charophytes have distinctive circular rings of proteins embedded in the plasma membrane **(Figure 29.2)**; these protein rings synthesize the cellulose found in the cell wall. In contrast, noncharophyte algae have linear sets of proteins that synthesize cellulose. Likewise, in species of plants that have flagellated sperm, the structure of the sperm closely resembles that of charophyte sperm. Analyses of nuclear, chloroplast, and mitochondrial DNA also support the close relationship between plants and charophytes.

Figure 29.2

Rings of cellulose-synthesizing proteins.

Rings of cellulose-synthesizing proteins on a 20 nanometer scale.

(inverted-contrast TEM)

More specifically, a recent analysis of nearly 900 protein-coding nuclear genes found that charophytes in the clade Zygnematophyceae—such as *Zygnema*—are the closest living relatives of plants **(Figure 29.3)**. Although this evidence shows that plants arose from within a group of charophyte algae, it does not mean that plants are descended from these living algae. Even so, present-day charophytes may tell us something about the algal ancestors of plants.

Figure 29.3

Zygnema, an alga closely related to plants.

Zygnema, an alga in a tube-like structure. Scale = 20 micrometers.

Zygnema is a filamentous pond alga in the Zygnematophyceae, the group of charophytes that is most closely related to plants.

Adaptations Enabling the Move to Land

Many species of charophyte algae inhabit shallow waters around the edges of ponds and lakes, where they are subject to occasional drying. In such environments, natural selection favors individual algae that can survive periods when they are not submerged. In charophytes, a layer of a durable polymer called **sporopollenin** prevents exposed zygotes from drying out. A similar chemical adaptation is found in the tough sporopollenin walls that encase plant spores.

The accumulation of such traits by at least one population of charophyte algae (now extinct) enabled their descendants—the first plants—to live permanently above the waterline. This ability opened a new frontier: a

terrestrial habitat that offered enormous benefits. The bright sunlight was unfiltered by water and plankton; the atmosphere offered more plentiful carbon dioxide than did water; and the soil by the water's edge was rich in some mineral nutrients. But these benefits were accompanied by challenges: a relative scarcity of water and a lack of structural support against gravity. (To appreciate why such support is important, picture how the soft body of a jellyfish (jelly) collapses when waves strand it on a beach.) Plants diversified as new adaptations arose that enabled them to thrive despite these challenges.

Today, what adaptations are unique to plants? The answer depends on where you draw the boundary dividing plants from algae **(Figure 29.4)**. Since the placement of this boundary is the subject of ongoing debate, this text uses a traditional definition that equates the kingdom Plantae with embryophytes (plants with embryos). In this context, let's now examine the derived traits that separate plants from their closest algal relatives.

Figure 29.4

Three possible "plant" kingdoms, the clades Viridiplantae, Streptophyta, and Plantae.

Tree diagram for the evolution of algae, chlorophytes, charophytes and embryophytes from ancestral alga.

Tree diagram for the evolution of algae, chlorophytes, charophytes and embryophytes from ancestral alga with limbs represented as a list with the top of the list. Common ancestors are identified at branch points if labeled; if not, branch point is noted.

Branch point 1: Ancestral alga Red algae.

Branch point 2 (Viridiplantae). Chlorophytes.

Branch Point 3 (Streptophyta). Other charophytes.

Branch point 4. Closest charophyte relative. Embryophytes (Plantae).

Derived Traits of Plants

Several adaptations that facilitate survival and reproduction on dry land emerged after plants diverged from their algal relatives. Examples of such traits that are found in plants but not in charophyte algae include the following:

- **Alternation of generations.** This type of life cycle, consisting of multicellular forms that give rise to each other in turn, is described in **Figure 29.5**.

Figure 29.5

Exploring Alternation of Generations

Exploring derived traits of land plants – alternation of generations, multicellular dependent embryos, walled spores produced in sporangia, multicellular gametangia, and apical meristems.

Charophyte algae lack the key traits of plants described in this figure: alternation of generations; multicellular, dependent embryos; walled spores produced in sporangia; multicellular gametangia; and apical meristems. This suggests that these traits were absent in the ancestor common to plants and charophytes but instead evolved as derived traits of plants. Not every plant exhibits all of these traits; certain lineages of plants have lost some traits over time. Five steps of alternation of generations from the gametophyte producing haploid gametes (mitosis), zygote formation (fertilization), sporophyte formation (mitosis), haploid spores (meiosis), and gametophytes (mitosis), and the cycle continues. The life cycles of all plants alternate between two generations of distinct multicellular organisms: gametophytes and sporophytes. As shown in the diagram below (using a fern as an example), each generation gives rise to the other, a process that is called alternation of generations. This type of reproductive cycle evolved in various groups of algae but does not occur in the charophytes, the algae most closely related to plants. Take care not to confuse the alternation of generations in plants with the haploid and diploid stages in the life cycles of other sexually reproducing organisms (see Figure 13.6). Alternation of generations is distinguished by the fact that the life cycle includes both multicellular haploid organisms and multicellular diploid organisms. The multicellular haploid gametophyte (gamete-producing plant) is named for its production by mitosis of haploid gametes—eggs and sperm—that fuse during fertilization, forming diploid zygotes. Mitotic division of the zygote produces a multicellular diploid sporophyte (spore-producing plant). Meiosis in a mature sporophytes produces haploid spores, reproductive cells that can develop into a new haploid organism without fusing with another cell. Mitotic division of the spore cell produces a new multicellular gametophyte, and the cycle begins again. Image: 1. The gametophyte (n) produces haploid gametes by mitosis. Gamete (n) from another plant is introduced and 2. The two gametes unite (fertilization) and form diploid zygote. 3.The zygote (2 n) develops into a multicellular diploid sporophyte by mitosis. 4. The sporophyte produces unicellular haploid spores by meiosis. 5. The spores (n) develop into multicellular haploid gametophytes and the cycle begins again. Key: haploid (n), diploid (2 n).

MAKE CONNECTIONS *Review sexual life cycles in*Figure 13.6*. Identify which type of sexual life cycle has alternation of generations, and summarize how it differs from other life cycles.*

*For suggested answer, see*Appendix A.*

- **Walled spores produced in sporangia.** The sporophyte stage of the plant life cycle has multicellular organs called **sporangia** (singular, *sporangium*) that produce spores **(Figure**

29.6). The polymer sporopollenin makes the walls of these spores resistant to harsh environments, enabling plant spores to be dispersed through dry air without harm.

Figure 29.6

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Sporophytes and sporangia of a moss in the genus *Mnium*.
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Sporophytes and sporangia of Sphagnum (a moss).

Each of the many spores produced by a sporangium is encased by a durable, sporopollenin-enriched wall.

- **Apical meristems**. Plants also differ from their algal relatives in having ********apical meristems**, localized regions of cell division at the tips of roots and shoots **(Figure 29.7)**. Apical meristem cells can divide throughout the plant's life, enabling its roots and shoots to elongate, thus increasing the plant's exposure to environmental resources.

Figure 29.7

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Apical meristem of a plant shoot.
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Apical meristems of plant roots and shoots. The LMs are longitudinal sections at the tips of a root and shoot.

Cells produced by apical meristems differentiate into the outer epidermis, which protects the body, and various types of internal tissues, such as those in leaves (LM).

Additional derived traits that relate to terrestrial life have evolved in many plant species. For example, the epidermis in many species has a covering, the **cuticle**, which consists of wax and other polymers. Permanently exposed to the air, plants run a far greater risk of desiccation (drying out) than do their algal relatives. The cuticle acts as waterproofing, helping prevent excessive water loss from the aboveground plant organs, while also providing some protection from microbial attack. Most plants also have specialized pores called **stomata** (singular, *stoma*), which support photosynthesis by allowing the exchange of CO2 and O2 between the outside air and the plant (see Figure 10.3). Stomata are also the main avenues by which water evaporates from the plant; in hot, dry conditions, the stomata close, minimizing water loss.

The earliest plants lacked true roots and leaves. Without roots, how did these plants absorb nutrients from the soil? Fossils dating from 420 million years ago reveal an adaptation that may have aided early plants in nutrient uptake: They formed symbiotic associations with fungi. We'll explore these associations, called

mycorrhizae, and their benefits to both plants and fungi in more detail in Concept 31.1. For now, the main point is that mycorrhizal fungi form extensive networks of filaments through the soil and transfer nutrients to their symbiotic plant partner. This benefit may have helped plants without roots to colonize land.

The Origin and Diversification of Plants

The algae most closely related to plants include many unicellular species and small colonial species. Since it is likely that the first plants were similarly small, the search for the earliest fossils of plants has focused on the microscopic world. As mentioned earlier, microorganisms colonized land as early as 3.2 billion years ago. But the microscopic fossils that document life on land changed dramatically 470 million years ago with the appearance of spores from early plants.

What distinguishes these spores from those of algae or fungi? One clue comes from their chemical composition, which matches the composition of plant spores today but differs from that of the spores of other organisms. In addition, the walls of these ancient spores have structural features that today are found only in the spores of certain plants (liverworts). And in rocks dating to 450 million years ago, researchers have discovered similar spores embedded in plant cuticle material that resembles spore-bearing tissue in living plants **(Figure 29.8)**.

Figure 29.8

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Ancient plant spores and tissue.
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Ancient plant spores and tissue.

(colorized SEMs)

Part a. Fossilized spores. The chemical composition and the physical structure of the walls of these 450-million-year-old spores match those found in certain plants today.

Part b. Fossilized sporophyte tissue. The spores in part a were embedded in tissue that appears to be from plants. Scale = 50 micrometers.

Fossils of larger plant structures, such as the *Cooksonia* sporangium in **Figure 29.9**, date to 425 million years ago, which is 45 million years after the appearance of plant spores in the fossil record. While the precise age (and form) of the first plants has yet to be discovered, those ancestral species gave rise to the vast diversity of living plants. **Table 29.1** provides basic information about the ten extant phyla. (Extant lineages are those that have surviving members.) As you read the rest of this section, look at Table

29.1 together with **Figure 29.10**, which reflects a view of plant phylogeny that is based on plant anatomy, biochemistry, and genetics.

Figure 29.9

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*Cooksonia* sporangium fossil.
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Mushroom-shaped fossil of Cooksonia sporangium; scale = 0.3 mm.

With the exception of the shape of its sporangia, *Cooksonia's* overall form was very similar to the early plant shown in Figures 29.1 and 29.16.

Table 29.1

Ten Phyla of Extant Plants

A table titled Table 29.1 Ten phyla of extant plants illustrates common name and number of known species.

	Common Name	Number of Known Species
Nonvascular Plants (Bryophytes)		
Phylum Hepatophyta	Liverworts	9,000
Phylum Bryophyta	Mosses	13,000
Phylum Anthocerophyta	Hornworts	225
Vascular Plants		
Seedless Vascular Plants		
Phylum Lycophyta	Lycophytes	1,200
Phylum Monilophyta	Monilophytes	12,000
Seed Plants		
Gymnosperms		
Phylum Ginkgophyta	Ginkgo	1
Phylum Cycadophyta	Cycads	350

Phylum Gnetophyta	Gnetophytes	75
Phylum Coniferophyta	Conifers	600
Angiosperms		
Phylum Anthophyta	Flowering plants	290,000

Figure 29.10

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Highlights of plant evolution.
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Tree diagram for the highlights of plant evolution showing land plants (nonvascular plants, seedless vascular plants, and seed plants) from ancestral green alga.

The phylogeny shown here illustrates a leading hypothesis about the relationships between plant groups.

Tree diagram for the highlights of plant evolution of land plants (nonvascular plants, seedless vascular plants, and seed plants) from ancestral green alga with limbs represented as a list.

Branch point 1: Ancestral green alga (origin of land plants (about 470 million years ago (million years ago)). Liverworts, nonvascular plants (bryophytes).

Branch point 2 (about 460 million years ago). Mosses - nonvascular plants (bryophytes).

Branch Point 3 (about 440 million years ago). Hornworts - nonvascular plants (bryophytes)

Branch Point 4, origin of vascular plants (about 425 million years ago). Lycophytes (club mosses, spike mosses, quillworts) - seedless vascular plants.

Branch point 5 (about 390 million years ago). Monilophytes (ferns, horsetails, whisk ferns), seedless vascular plants.

Branch point 6, origin of seed plants (about 360 million years ago). Gymnosperms. Angiosperms.

MAKE CONNECTIONS *The figure identifies which lineages are plants, nonvascular plants, vascular plants, seedless vascular plants, and seed plants. Which of these categories are monophyletic, and which are paraphyletic? Explain. (See**Figure 26.10**to review these terms.)*

*For suggested answer, see*Appendix A.**

One way to distinguish groups of plants is whether or not they have an extensive system of **vascular tissue**, cells joined into tubes that transport water and nutrients throughout the plant body. Most present-day

plants have a complex vascular tissue system and are therefore called **vascular plants**. Plants that do not have an extensive transport system—liverworts, mosses, and hornworts—are described as "nonvascular" plants, even though some mosses do have simple vascular tissue. Nonvascular plants are often informally called **bryophytes** (from the Greek *bryon*, moss, and *phyton*, plant). Although the term *bryophyte* is commonly used to refer to all nonvascular plants, molecular studies and morphological analyses of sperm structure have concluded that bryophytes do not form a monophyletic group (a clade).

Vascular plants, which form a clade that comprises about 93% of all extant plant species, can be categorized further into smaller clades. Two of these clades are the **lycophytes** (the club mosses and their relatives) and the **monilophytes** (ferns and their relatives). The plants in each of these clades lack seeds, which is why collectively the two clades are informally called **seedless vascular plants**. However, notice in Figure 29.10 that, like bryophytes, seedless vascular plants do not form a clade.

A group such as the bryophytes or the seedless vascular plants represents a collection of organisms that share key biological features. It can be informative to group organisms according to their features, such as having a vascular system but lacking seeds. But members of such a group, unlike members of a clade, do not necessarily share the same ancestry. For example, even though monilophytes and lycophytes are all seedless vascular plants, monilophytes share a more recent common ancestor with seed plants. As a result, we would expect monilophytes and seed plants to share key traits not found in lycophytes—and they do, as you'll read in Concept 29.3.

A third clade of vascular plants consists of seed plants, which represent the vast majority of living plant species. A **seed** is an embryo packaged with a supply of nutrients inside a protective coat. Seed plants can be divided into two groups, gymnosperms and angiosperms, based on the absence or presence of enclosed chambers in which seeds mature. **Gymnosperms** (from the Greek *gymnos*, naked, and *sperm*, seed) are known as "naked seed" plants because their seeds are not enclosed in chambers. Living gymnosperm species, the most familiar of which are the conifers, form a clade. **Angiosperms** (from the Greek *angion*, container) are a huge clade consisting of all flowering plants; their seeds develop inside chambers that originate within flowers. Nearly 90% of living plant species are angiosperms.

Note that the phylogeny depicted in Figure 29.10 focuses only on the relationships between extant plant lineages. Paleobotanists have also discovered fossils belonging to extinct plant lineages. As you'll read later in the chapter, these fossils can reveal intermediate steps in the emergence of plant groups found on Earth today.

Concept 29.2: Mosses and other nonvascular plants have life cycles dominated by gametophytes

An evolutionary tree diagram highlights nonvascular plants (bryophytes).

Branch point 1: First level-1 branch: Nonvascular plants (bryophytes). The second level-1 branch splits from branch point 2 into the first level-2 branch: seedless vascular plants. The second level-2 branch splits from branch point 3 into the first level-3 branch: Gymnosperms and the second level-3 branch: Angiosperms.

The nonvascular plants (bryophytes) are represented today by three phyla of small, herbaceous (nonwoody) plants: **liverworts** (phylum Hepatophyta), **mosses** (phylum Bryophyta), and **hornworts** (phylum Anthocerophyta). Liverworts and hornworts are named for their shapes, plus the suffix *wort* (from the Anglo-Saxon for “herb”). Mosses are familiar to many people, although some plants commonly called “mosses” are not really mosses at all. These include Irish moss (a red seaweed), reindeer moss (a lichen), club mosses (seedless vascular plants), and Spanish mosses (lichens in some regions and flowering plants in others).

Phylogenetic analyses indicate that liverworts, mosses, and hornworts diverged from other plant lineages early in the history of plant evolution (see Figure 29.10). Fossil evidence provides some support for this idea: The earliest spores of plants (dating from 450 to 470 million years ago) have structural features found only in the spores of liverworts, and by 430 million years ago spores similar to those of mosses and hornworts also occur in the fossil record. The earliest fossils of vascular plants date to about 425 million years ago.

Over the long course of their evolution, liverworts, mosses, and hornworts have acquired many unique adaptations. Next, we’ll examine some of those features.

Bryophyte Gametophytes

Unlike vascular plants, in all three bryophyte phyla the haploid gametophytes are the dominant stage of the life cycle: They are usually larger and longer-living than the sporophytes, as shown in the moss life cycle in **Figure 29.11**. The sporophytes are typically present only part of the time.

Figure 29.11

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The life cycle of a moss.
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Life cycle of a moss showing the male and female gametophyte.

The cycle is as follows.

1.Spores develop into threadlike protonemata (n).

2.The haploid protonemata produce buds that divide by mitosis and grow into gametophytes. Male gametophyte (n) have antheridia that produce many sperm. Female gametophyte (n) have archegonia that each produce one egg.

3.Sperm must swim through a film of moisture to reach the egg and then fertilization occurs with the archegonium. The Zygote (2 n) forms in the archegonium and

4.develops into a sporophyte embryo.

5.The sporophyte grows a long stalk (seta) that emerges from the archegonium. A capsule (sporangium) forms at the end of the seta.

6.Attached by its foot, the young sporophyte (2 n) remains nutritionally dependent on the gametophyte.

7.Meiosis occurs and haploid spores develop in the capsule. When the capsule is mature, its lid pops off, and the spores are released through the peristome.

VISUAL SKILLS *In this diagram, does the sperm cell that fertilizes the egg cell differ genetically from the egg? Explain.*

*For suggested answer, see*Appendix A.**

Animation

Moss Life Cycle

image

Play

image

Play

When bryophyte spores are dispersed to a favorable habitat, such as moist soil or tree bark, they may germinate and grow into gametophytes. Germinating moss spores, for example, characteristically produce a mass of green, branched, one-cell-thick filaments known as a **protonema** (plural, *protonemata*). A

protonema has a large surface area that enhances absorption of water and minerals. In favorable conditions, a protonema produces one or more “buds,” each of which develops into a moss gametophyte. (Note that when referring to nonvascular plants, we often use quotation marks for structures similar to the buds, stems, and leaves of vascular plants because the definitions of these terms are based on vascular plant organs.)

Bryophyte gametophytes generally form ground-hugging carpets, partly because their body parts are too thin to support a tall plant. A second constraint on the height of many bryophytes is the absence of vascular tissue, which would enable long-distance transport of water and nutrients. (The thin structure of bryophyte organs makes it possible to distribute materials for short distances without specialized vascular tissue.) However, some mosses have conducting tissues in the center of their “stems.” A few of these mosses can grow as tall as 60cm 2 feet as a result. Phylogenetic analyses suggest that conducting tissues similar to those of vascular plants arose independently in these mosses by convergent evolution.

The gametophytes are anchored to the ground by delicate **rhizoids**, which are long, tubular single cells (in liverworts and hornworts) or filaments of cells (in mosses). Unlike roots, which are found in vascular plant sporophytes, rhizoids lack specialized conducting cells and do not play a primary role in water and mineral absorption.

Gametophytes can form multiple****__gametangia__, multicellular structures that produce gametes and are covered by protective tissue. Female gametangia are called **archegonia**, and male gametangia are called **antheridia**. Each archegonium produces one egg, whereas each antheridium produces many sperm. Some bryophyte gametophytes are bisexual, but in mosses the archegonia and antheridia are typically carried on separate female and male gametophytes. Flagellated sperm swim through a film of water toward eggs, entering the archegonia in response to chemical attractants. Eggs are not released but instead remain within the bases of archegonia. After fertilization, embryos are retained within the archegonia. Layers of placental transfer cells help transport nutrients to the embryos as they develop into sporophytes.

Bryophyte sperm typically require a film of water to reach the eggs. Given this requirement, it is not surprising that many bryophyte species are found in moist habitats. The fact that sperm swim through water to reach the egg also means that in species with separate male and female gametophytes (most species of mosses), sexual reproduction is likely to be more successful when individuals are located close to one another.

Many bryophyte species can increase the number of individuals in a local area through various methods of asexual reproduction. For example, some mosses reproduce asexually by forming *brood bodies*, small plantlets that detach from the parent plant and grow into new, genetically identical copies of their parent **(Figure 29.12)**.

Figure 29.12

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A moss brood body.
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A moss brood body has a stem-like body with a tree-like top.

Bryophyte Sporophytes

The cells of bryophyte sporophytes contain plastids that are usually green and photosynthetic when the sporophytes are young. Even so, bryophyte sporophytes cannot live independently. A bryophyte sporophyte remains attached to its parental gametophyte throughout the sporophyte's lifetime, dependent on the gametophyte for supplies of sugars, amino acids, minerals, and water.

Bryophytes have the smallest sporophytes of all extant plant groups. A typical bryophyte sporophyte consists of a foot, a seta, and a sporangium. Embedded in the archegonium, the **foot** absorbs nutrients from the gametophyte. The **seta** (plural, *setae*), or stalk, conducts these materials to the sporangium, also called a **capsule**, which produces spores by meiosis.

Bryophyte sporophytes can produce enormous numbers of spores. A single moss capsule, for example, can generate over 5 million spores. In most mosses, the seta becomes elongated, enhancing spore dispersal by elevating the capsule. Typically, the upper part of the capsule features a ring of interlocking, tooth-like structures known as the **peristome** (see Figure 29.11). These "teeth" open under dry conditions and close again when it is moist. This allows moss spores to be discharged gradually, via periodic gusts of wind that can carry them long distances.

Moss and hornwort sporophytes are often larger and more complex than those of liverworts. For example, hornwort sporophytes, which superficially resemble grass blades, have a cuticle. Moss and hornwort sporophytes also have stomata, as do all vascular plants (but not liverworts).

Figure 29.13 shows some examples of gametophytes and sporophytes in the bryophyte phyla.

Figure 29.13

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Exploring Bryophyte Diversity
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Exploring bryophyte diversity.

Three plants are described as follows.

Liverworts (phylum Hepatophyta). This phylum's common and scientific names (from the Latin hepaticus, liver) refer to the liver-shaped gametophytes of its members, such as Marchantia, shown in a. In medieval

times, their shape was thought to be a sign that the plants could help treat liver diseases. Some liverworts, including Marchantia, are described as thalloid because of the flattened shape of their gametophytes. Marchantia gametangia are elevated on gametophores that look like miniature trees. You would need a magnifying glass to see the sporophytes, which have a short seta (stalk) with an oval or round capsule. Other liverworts, such as Plagiochila, shown in b, are called leafy because their stem-like gametophytes have many leaflike appendages. There are many more species of leafy liverworts than thalloid liverworts.

Hornworts (phylum Anthocerophyta). This phylum's common and scientific names (from the Greek keras, horn) refer to the long, tapered shape of the sporophyte. A typical sporophyte can grow to about 5 cm high. Unlike a liverwort or moss sporophyte, a hornwort sporophyte lacks a seta and consists only of a sporangium. The sporangium releases mature spores by splitting open, starting at the tip of the horn. The gametophytes, which are usually 1–2 centimeters in diameter, grow mostly horizontally and often have multiple sporophytes attached. Hornworts are frequently among the first species to colonize open areas with moist soils; a symbiotic relationship with nitrogen-fixing cyanobacteria contributes to their ability to do this (nitrogen is often in short supply in such areas).

The Ecological and Economic Importance of Mosses

Wind dispersal of lightweight spores has distributed mosses throughout the world. These plants are particularly common and diverse in moist forests and **wetlands**. Some mosses colonize bare, sandy soil, where, researchers have found, they help retain nitrogen in the soil **(Figure 29.14)**. In northern coniferous forests, species such as the feather moss *Pleurozium* harbor nitrogen-fixing cyanobacteria that increase the availability of nitrogen in the ecosystem. Other mosses inhabit such extreme environments as mountaintops, **tundra**, and deserts. Many mosses are able to live in very cold or dry habitats because they can survive the loss of most of their body water, then rehydrate when moisture is available. Few vascular plants can survive the same degree of desiccation. Moreover, phenolic compounds in moss cell walls absorb damaging levels of UV radiation present in deserts or at high altitudes.

Figure 29.14

Inquiry: Can bryophytes reduce the rate at which key nutrients are lost from soils?

Experiment

Soils in terrestrial ecosystems are often low in nitrogen, a nutrient required for normal plant growth. Richard Bowden, of Allegheny College, measured annual inputs (gains) and outputs (losses) of nitrogen in a sandy soil ecosystem dominated by the moss *Polytrichum*. Nitrogen inputs were measured from rainfall (dissolved ions, such as nitrate, NO_3^-), biological N_2 fixation, and wind deposition. Nitrogen losses were measured in leached water (dissolved ions, such as NO_3^-) and gaseous emissions (such as N_2O emitted by bacteria). Bowden measured losses for soils with *Polytrichum* and for soils where the moss was removed two months before the experiment began.

Results

A total of 10.5kg of nitrogen per hectare (kg/ha) entered the ecosystem each year. Little nitrogen was lost by gaseous emissions 0.10 kg/ha·yr. The results of comparing nitrogen losses by leaching are shown below.

A bar graph plots annual nitrogen loss versus moss. A bar for, with moss, rises to approximately 0.40, and the bar for, without moss, rises to approximately 5.2.

Conclusion

The moss *Polytrichum* greatly reduced the loss of nitrogen by leaching in this ecosystem. Each year, the moss ecosystem retained over 95% of the 10.5kg/ha of total nitrogen inputs (only 0.1kg/ha and 0.3kg/ha were lost to gaseous emissions and leaching, respectively).

Data from R. D. Bowden, Inputs, outputs, and accumulation of nitrogen in an early successional moss (*Polytrichum*) ecosystem, *Ecological Monographs* 61:207–223 (1991).

WHAT IF?***How might the presence of* Polytrichum *affect plant species that typically colonize the sandy soils after the moss?*

*For suggested answer, see*Appendix A.**

One wetland moss genus, *Sphagnum*, or peat moss, is often a major component of deposits of partially decayed organic material known as **peat(Figure 29.15a)**. Boggy regions with thick layers of peat are called peatlands. *Sphagnum* does not decay readily, in part because of phenolic compounds embedded in its cell walls. The low temperature, pH, and oxygen level of peatlands also inhibit decay of moss and other organisms in these boggy wetlands. As a result, some peatlands have preserved corpses for thousands of years **(Figure 29.15b)**.

Figure 29.15

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    *Sphagnum*, or peat moss: a bryophyte with economic,
ecological, and archaeological significance.
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Part a. Peat being harvested from a peatland. Part b. Tollund Man, a bog mummy dating from 405-100 B.C.E. The acidic, oxygen-poor conditions produced by Sphagnum can preserve human or other animal bodies for thousands of years.

Peat has long been a fuel source in Europe and Asia, and it is still harvested for fuel today, notably in Ireland and Canada. Peat moss is also useful as a soil conditioner and for packing plant roots during shipment because it has large dead cells that can absorb roughly 20 times the moss's weight in water.

Peatlands cover just 3% of Earth's land surface but contain one-third of the world's soil carbon: Globally, an estimated 500 billion tons of organic carbon is stored as peat. Current overharvesting of *Sphagnum*—primarily for use in peat-fired power stations—contributes to global warming by releasing stored CO2. In addition, if global temperatures continue to rise, the water levels of some peatlands are expected to drop. Such a change would expose peat to air and cause it to decompose, thereby releasing additional stored CO2 and contributing further to global warming. The historical and expected future effects of *Sphagnum* on the global climate underscore the importance of preserving and managing peatlands.

Mosses may have a long history of affecting climate change. In the **Scientific Skills Exercise**, you will explore the question of whether they did so during the Ordovician period by contributing to the weathering of rocks.

Scientific Skills Exercise

Making Bar Graphs and Interpreting Data

AP Science Practices 1, 3, 4

Could Nonvascular Plants Have Caused Weathering of Rocks and Contributed to Climate Change During the Ordovician Period?The oldest traces of terrestrial plants are fossilized spores formed 470 million years ago. Between that time and the end of the Ordovician period 444 million years ago, the atmospheric CO2 level dropped by half, and the climate cooled dramatically.

One possible cause of the drop in CO2 during the Ordovician period is the breakdown, or weathering, of rock. As rock weathers, calcium silicate (Ca2SiCO3) is released and combines with CO2 from the air,

producing calcium carbonate ($CaCO_3$). Today, the roots of vascular plants increase rock weathering and mineral release by producing acids that break down rock and soil. Although nonvascular plants lack roots, they require the same mineral nutrients as vascular plants. Could nonvascular plants also increase the chemical weathering of rock? If so, they could have contributed to the decline in atmospheric CO_2 during the Ordovician. In this exercise, you will interpret data from a study of the effects of moss on releasing minerals from two types of rock.

Fossilized spores.

How the Experiment Was Done The researchers set up experimental and control microcosms, or small artificial ecosystems, to measure mineral release from rocks. First, they placed rock fragments of volcanic origin, either granite or andesite, into small glass containers. Then they mixed water and macerated (chopped and crushed) moss of the species *Physcomitrella patens*. They added this mixture to the experimental microcosms (72 granite and 41 andesite). For the control microcosms (77 granite and 37 andesite), they filtered out the moss and just added the water. After 130 days, they measured the amounts of various minerals found in the water in the control microcosms and in the water and moss in the experimental microcosms.

Data from the Experiment The moss grew (increased its biomass) in the experimental microcosms. The table shows the mean amounts in micromoles (μmol) of several minerals measured in the water and the moss in the microcosms.

A table illustrates values of granite and andesite for calcium, magnesium, and potassium ions.

Data from T. M. Lenton et al., First plants cooled the Ordovician, *Nature Geoscience* 5:86–89 (2012).

	C a superscript 2 plus (micromoles)	**M g superscript 2 plus (micromoles)**	**K superscript plus (micromoles)**
Granite	Andesite	Granite	Andesite
Mean weathered amount released in water in the control microcosms	1.68	1.54	0.42
Mean weathered amount released in water in the experimental microcosms	1.27	1.84	0.34

Mean weathered amount taken up by moss in the experimental microcosms	1.09	3.62	0.31

INTERPRET THE DATA

- Why did the researchers add filtrate from which macerated moss had been removed to the control microcosms?
- Make two bar graphs (for granite and andesite) comparing the mean amounts of each element weathered from rocks in the control and experimental microcosms. (Hint: For an experimental microcosm, what sum represents the total amount weathered from rocks?) (For additional information on bar graphs, see the Scientific Skills Review in Appendix D.)
- Overall, what is the effect of moss on chemical weathering of rock? Are the results similar or different for granite and andesite?
- Based on their experimental results, the researchers added weathering of rock by nonvascular plants to simulation models of the Ordovician climate. The new models predicted decreased CO_2 levels and global cooling sufficient to produce the glaciations in the late Ordovician period. What assumptions did the researchers make in using results from their experiments in climate simulation models?
- "Life has profoundly changed the Earth." Explain whether or not these experimental results support this statement.

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

Concept 29.3: Ferns and other seedless vascular plants were the first plants to grow tall

On an evolutionary tree, seedless vascular plants are highlighted.

Branch point 1: First level-1 branch: Nonvascular plants (bryophytes). The second level-1 branch splits from branch point 2 into the first level-2 branch: seedless vascular plants. The second level-2 branch splits from branch point 3 into the first level-3 branch: Gymnosperms and the second level-3 branch: Angiosperms.

During the first 100 million years of plant evolution, bryophytes were prominent types of vegetation. But it is vascular plants that dominate most landscapes today. The earliest fossils of vascular plants date to 425 million years ago. These plants lacked seeds but had well-developed vascular systems, an evolutionary novelty that set the stage for vascular plants to grow taller than their bryophyte counterparts. As in bryophytes, however, the sperm of ferns and all other seedless vascular plants are flagellated and swim through a film of water to reach eggs. In part because of these swimming sperm, seedless vascular plants today are most common in damp environments.

Origins and Traits of Vascular Plants

Unlike the nonvascular plants, ancient relatives of vascular plants had branched sporophytes that were not dependent on gametophytes for nutrition **(Figure 29.16)**. Although these early plants were less than 20 cm tall, their branching enabled their bodies to become more complex and to have multiple sporangia. As plant bodies became more complex over time, competition for space and sunlight probably increased. As we'll see, that competition may have stimulated still more evolution in vascular plants, eventually leading to the formation of the first forests.

Figure 29.16

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    Sporophytes of *Aglaophyton major*, an ancient relative of
living vascular plants.
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Sporophytes of Aglaophyton major.

This reconstruction from 405-million-year-old fossils exhibits dichotomous (Y-shaped) branching with sporangia at the ends of branches. Sporophyte branching characterizes living vascular plants but is lacking in living nonvascular plants (bryophytes). *Aglaophyton* had structures called rhizoids that anchored it to the ground. The inset shows a fossilized stoma of *A. major* (colorized LM).

Early vascular plants had some derived traits of today's vascular plants, but they lacked roots and some other adaptations that evolved later. The main traits that characterize living vascular plants are life cycles with dominant sporophytes, transport in vascular tissues called xylem and phloem, and well-developed roots and leaves, including spore-bearing leaves called sporophylls.

Life Cycles with Dominant Sporophytes

As mentioned earlier, mosses and other bryophytes have life cycles dominated by gametophytes (see Figure 29.11). Fossil evidence suggests that a change began to develop in some of the earliest vascular plants, whose gametophytes and sporophytes were about equal in size. Further reductions in gametophyte size occurred among extant vascular plants; in these groups, the sporophyte generation is the larger and more complex form in the alternation of generations **(Figure 29.17)**. In ferns, for example, the familiar leafy plants are the sporophytes. You would have to get down on your hands and knees and search the ground carefully to find fern gametophytes, which are tiny structures that often grow on or just below the soil surface.

Figure 29.17

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The life cycle of a fern.
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The life cycle of a fern showing 1, sporangia releasing spores; 2, gametophyte developing reproducing organs; 3, fertilization, 4, zygote development, and 5, meiosis and spore dispersal.

1. Sporangia release spores. Most fern species produce a single type of spore (n) that develops into a bisexual photosynthetic gametophyte.
2. Each gametophyte develops sperm-producing organs called antheridia and egg-producing organs called archegonia. Although this simplified diagram shows a sperm fertilizing an egg from the same gametophyte, in most fern species a gametophyte produces sperm and eggs at different times. Therefore, typically an egg from one gametophyte is fertilized by a sperm from another gametophyte.
3. Sperm use flagella to swim to eggs in the archegonia. An attractant secreted by archegonia helps direct the sperm. Fertilization then occurs.
4. A zygote (2 n) develops into a new sporophyte, and the young plant grows out from an archegonium of its parent, the gametophyte. A mature sporophyte (2 n) and fiddlehead (young leaf) shown.
5. On the underside of the sporophyte's reproductive leaves are spots called sori. Each sorus is a cluster of sporangia. Meiosis occurs and the cycle continues when the sporangia release spores.

WHAT IF? *If the ability to disperse sperm by wind evolved in a fern, how might its life cycle be affected?*

*For suggested answer, see*Appendix A.**

Animation

Fern Life Cycle

image

Transport in Xylem and Phloem

Vascular plants have two types of vascular tissue: xylem and phloem (see Figure 35.10). **Xylem** conducts most of the water and minerals. The xylem of all vascular plants includes **tracheids**, tube-shaped cells that carry water and minerals up from the roots. The water-conducting cells of the xylem are dead at functional maturity and are *lignified*; that is, their cell walls are strengthened by the polymer **lignin**. The tissue called **phloem** has cells arranged into tubes that distribute sugars, amino acids, and other organic products; these cells are alive at functional maturity.

Lignified vascular tissue helped enable vascular plants to grow tall. Their stems became strong enough to provide support against gravity, and they could transport water and mineral nutrients high above the ground. Tall plants could also outcompete short plants for access to the sunlight needed for photosynthesis. In addition, the spores of tall plants could disperse farther than those of short plants, enabling tall species to colonize new environments more rapidly. Overall, the ability to grow tall gave vascular plants a competitive edge over nonvascular plants, which typically are less than 5cm in height. Over time, competition among vascular plants also would have increased, leading to selection for taller growth forms—a process that eventually gave rise to the trees that formed the first forests 385 million years ago.

Evolution of Roots

Vascular tissue also provides benefits below ground. Instead of the rhizoids seen in bryophytes, roots evolved in the sporophytes of almost all vascular plants. **Roots** are organs that absorb water and nutrients from the soil. Roots also anchor vascular plants to the ground, hence allowing the shoot system to grow taller.

Root tissues of living plants closely resemble stem tissues of early vascular plants preserved in fossils. This suggests that roots may have evolved from the lowest belowground portions of stems in ancient vascular plants. It is unclear whether roots evolved only once in the common ancestor of all vascular plants or independently in different lineages. Although the roots of living members of these lineages of vascular plants share many similarities, fossil evidence hints at convergent evolution. The oldest fossils of lycophytes, for example, already displayed simple roots 400 million years ago, when the ancestors of ferns and seed plants still had none. Studying genes that control root development in different vascular plant species may help resolve this question.

Evolution of Leaves

Leaves are structures that serve as the primary photosynthetic organ of vascular plants. In terms of size and complexity, leaves can be classified as either microphylls or megaphylls **(Figure 29.18)**. All of the lycophytes—and only the lycophytes—have **microphylls**, which are small, often spine-shaped leaves supported by a single strand of vascular tissue. Almost all other vascular plants have **megaphylls**, leaves with a highly branched vascular system; a few species have reduced leaves that appear to have evolved from megaphylls. Megaphylls are typically larger than microphylls and therefore support greater photosynthetic productivity than microphylls. Microphylls first appear in the fossil record 410 million years ago, but megaphylls do not emerge until 370 million years ago, toward the end of the Devonian period.

Figure 29.18

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Microphyll and megaphyll leaves.
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Microphyll leaves with unbranched vascular tissue (inset Selaginella kraussiana, Krauss's spike moss). Megaphyll leaves with branched vascular tissue (inset Hymenophyllum tunbrigense, Tunbridge filmy fern).

Sporophylls and Spore Variations

One milestone in the evolution of plants was the emergence of **sporophylls**, modified leaves that bear sporangia. Sporophylls vary greatly in structure. For example, fern sporophylls produce clusters of sporangia known as **sori** (singular, *sorus*), usually on the undersides of the sporophylls (see Figure 29.17). In many lycophytes and in most gymnosperms, groups of sporophylls form cone-like structures called **strobili** (singular, *strobilus*). The sporophylls of angiosperms are called *carpels* and *stamens* (see Figure 30.8).

Most seedless vascular plant species are **homosporous**: They have one type of sporophyll bearing one type of sporangium that produces one type of spore, which typically develops into a bisexual gametophyte, as in most ferns. In contrast, a **heterosporous** species has two types of sporophylls, called megasporophylls and microsporophylls. Megasporophylls have megasporangia, which produce **megaspores**, spores that develop into female gametophytes. Microsporophylls have microsporangia, which produce **microspores**, smaller spores that develop into male gametophytes. All seed plants and a few seedless vascular plants are heterosporous. The following diagram compares the two conditions:

Two charts for spore production.

Homosporous spore production (most seedless vascular plants): Sporangium on sporophyll goes to a single type of spore goes to typically a bisexual gametophyte which produces eggs and sperm.

Heterosporous spore production (all seed plants): Megasporangium on megasporophyll goes to megaspore goes to female gametophyte goes to eggs. Microsporangium on microsporophyll goes to microspore goes to male gametophyte goes to sperm.

Classification of Seedless Vascular Plants

As we noted earlier, biologists recognize two clades of living seedless vascular plants: the lycophytes (phylum Lycophyta) and the monilophytes (phylum Monilophyta). The lycophytes include the club mosses, the spikemosses, and the quillworts. The monilophytes include the ferns, the horsetails, and the whisk ferns and their relatives. Although ferns, horsetails, and whisk ferns differ greatly in appearance, recent anatomical and molecular comparisons provide convincing evidence that these three groups make up a clade. Accordingly, many systematists now classify them together as the phylum Monilophyta, as we do in this chapter. Others refer to these groups as three separate phyla within a clade. **Figure 29.19** describes the two main groups of seedless vascular plants.

Figure 29.19

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Exploring Seedless Vascular Plant Diversity
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Exploring seedless vascular plant diversity.

Four plants are described as follows.

Lycophytes (phylum Lycophyta). Many lycophytes grow on tropical trees as epiphytes, plants that use other plants as a substrate but are not parasites. Other species grow on temperate forest floors. In some species, the tiny gametophytes live above ground and are photosynthetic. Others live below ground, nurtured by symbiotic fungi. Sporophytes have upright stems with many small leaves, as well as ground-hugging stems that produce dichotomously branching roots. Spike mosses are usually relatively small and often grow horizontally. In many club mosses and spike mosses, sporophylls are clustered into club-shaped cones (strobili). Quillworts, named for their leaf shape, form a single genus whose members live in marshy areas or as submerged aquatic plants. Club mosses are all homosporous, whereas spike mosses and quillworts are all heterosporous. The spores of club mosses are released in clouds and are so rich in oil that magicians and photographers once ignited them to create smoke or flashes of light.

Monilophytes (phylum Monilophyta) include ferns, horsetails, and whisk ferns and relatives.

Ferns. Unlike the lycophytes, ferns have megaphylls (see Figure 29.13). The sporophytes typically have horizontal stems that give rise to large leaves called fronds, often divided into leaflets. A frond grows as its

coiled tip, the fiddlehead, unfurls. Almost all species are homosporous. The gametophyte in some species shrivels and dies after the young sporophyte detaches itself. In most species, sporophytes have stalked sporangia with spring like devices that catapult spores several meters. Airborne spores can be carried far from their origin. Some species produce more than a trillion spores in a plant's lifetime.

Horsetails. The group's name refers to the brushy appearance of the stems, which have a gritty texture that made them historically useful as scouring rushes for pots and pans. Some species have separate fertile (cone-bearing) and vegetative stems. Horsetails are homosporous, with cones releasing spores that typically give rise to bisexual gametophytes. Horsetails are also called arthrophytes (jointed plants) because their stems have joints. Rings of small leaves or branches emerge from each joint, but the stem is the main photosynthetic organ. Large air canals carry oxygen to the roots, which often grow in waterlogged soil.

Whisk ferns and relatives. Like primitive vascular plant fossils, the sporophytes of whisk ferns (genus Psilotum) have dichotomously branching stems but no roots. Stems have scale like outgrowths that lack vascular tissue and may have resulted from the evolutionary reduction of leaves. Each yellow knob on a stem consists of three fused sporangia. Species of the genus Tmesipteris closely related to whisk ferns and found only in the South Pacific, also lack roots but have small, leaflike outgrowths in their stems, giving them a vine-like appearance. Both genera are homosporous, with spores giving rise to bisexual gametophytes that grow underground and are only about a centimeter long.

Phylum Lycophyta: Club Mosses, Spikemosses, and Quillworts

Present-day species of lycophytes are relicts of a far more impressive past. By the Carboniferous period (359–299 million years ago), the lycophyte evolutionary lineage included small herbaceous plants and giant trees with diameters of more than 2m and heights of more than 40 m. The giant lycophyte trees thrived for millions of years in moist swamps, but their diversity declined when Earth's climate became drier during the Permian period (299–252 million years ago). The small lycophytes survived, represented today by about 1,200 species. Though some are commonly called club mosses and spikemosses, they are not true mosses (which, as discussed earlier, are nonvascular plants).

Phylum Monilophyta: Ferns, Horsetails, and Whisk Ferns and Relatives

Ferns radiated extensively from their Devonian origins and grew alongside lycophyte trees and horsetails in the great Carboniferous swamp forests. Today, ferns are by far the most widespread seedless vascular plants, numbering more than 12,000 species. Though most diverse in the tropics, many ferns thrive in temperate forests, and some species are even adapted to arid habitats.

As mentioned earlier, ferns and other monilophytes are more closely related to seed plants than to lycophytes. As a result, monilophytes and seed plants share traits that are not found in lycophytes, including megaphyll leaves and roots that can branch at various points along the length of an existing root. In lycophytes, by contrast, roots branch only at the growing tip of the root, forming a Y-shaped structure.

The monilophytes called horsetails were very diverse during the Carboniferous period, some growing as tall as 15 m. Today, only 15 species survive as a single, widely distributed genus, *Equisetum*, often found in marshy places and along streams.

Psilotum (whisk ferns) and a closely related genus, *Tmesipteris*, form a clade consisting mainly of tropical epiphytes. Plants in these two genera, the only vascular plants lacking true roots, once were called “living fossils” because of their resemblance to fossils of ancient relatives of living vascular plants (see Figures 29.16 and 29.19). However, much evidence, including analyses of DNA sequences and sperm structure, indicates that the genera *Psilotum* and *Tmesipteris* are closely related to ferns. This hypothesis suggests that their ancestor’s true roots were lost during evolution. Today, plants in these two genera absorb water and nutrients through numerous absorptive rhizoids.

The Significance of Seedless Vascular Plants

The ancestors of living lycophytes, horsetails, and ferns, along with their extinct seedless vascular relatives, grew to great heights during the Devonian and early Carboniferous, forming the first forests **(Figure 29.20)**. How did their dramatic growth affect Earth and its other life?

Figure 29.20

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    Artist’s conception of a Carboniferous forest based on
fossil evidence.
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Artist's conception of a Carboniferous forest.

Lycophyte trees, with trunks covered with small leaves, thrived in the “coal forests” of the Carboniferous, along with giant ferns and horsetails.

One major effect was that early forests contributed to a large drop in CO_2 levels during the Carboniferous period, causing global cooling that resulted in widespread glacier formation. The trees of early forests contributed to this drop in CO_2 levels in part by the actions of their roots. The roots of vascular plants secrete acids that break down rocks, thereby increasing the rate at which calcium and magnesium are released from rocks into the soil. These chemicals react with carbon dioxide dissolved in rain water, forming

compounds that ultimately wash into the oceans, where they are incorporated into rocks (calcium or magnesium carbonates). The net effect of these processes—which were accelerated by plants—is that CO_2 removed from the air is stored in marine rocks. Although carbon stored in these rocks can be returned to the atmosphere, it typically takes millions of years for this to occur (as when geological uplift brings the rocks to the surface, exposing them to erosion).

In addition, the seedless vascular plants that formed the first forests eventually became coal, again removing CO_2 from the atmosphere for long periods of time. In the stagnant waters of Carboniferous swamps, the dead bodies of early trees did not completely decay. This organic material turned to thick layers of peat, later covered by the sea. Marine sediments piled on top, and over millions of years, heat and pressure converted the peat to coal. In fact, Carboniferous coal deposits are the most extensive ever formed. Coal was crucial to the Industrial Revolution, and people worldwide still burn 6 billion tons a year. It is ironic that coal, formed from plants that contributed to a global cooling, now contributes to global warming by returning carbon to the atmosphere (see Figure 56.29).

Growing along with the seedless plants in Carboniferous swamps were primitive seed plants. Though seed plants were not dominant at that time, they rose to prominence after the swamps began to dry up at the end of the Carboniferous period. The next chapter traces the origin and diversification of seed plants, continuing our story of adaptation to life on land.